



M002 Extension 3 — Build Your Own Maze (Beginner)

Topic: Make a Maze + Write the Solution · **Course:** Maze Madness · **Level:** Extension (Beginner) ·

Time: about 45 minutes

This time **you** are the maze maker! You will build one long maze that goes through **6 parts**, add some pistons and doors, and then write code to get the agent to the end.

Open the **M002 EXT 3 — Cube World** and build inside it. Drive the agent with chat: `l` turns left, `r` turns right, `run` starts your code, `r1` sends the agent back to start.

Your maze needs at least:

- **2 pistons** (open a wall when powered)
- **2 doors** (the agent opens them)
- **4 AND** checks in your code (`... and ...`)
- **1 OR** check in your code (`... or ...`)

1 · Plan Your 6 Parts

Your maze has **6 parts** in a row. The agent finishes one part and walks into the next.

Draw or write what each part has. Easy idea: part = one room with one turn, one door, or one piston.

Part 1: ...

Part 2: ...

Part 3: ...

Part 4: ...

Part 5: ...

Part 6: ...

Which 2 parts will have a piston? Which 2 will have a door?

2 · Build It

Build your maze in the world. Walk it yourself first to make sure you can reach the end. Check off each part as you build it.

☐ Part 1 ☐ Part 2 ☐ Part 3 ☐ Part 4 ☐ Part 5 ☐ Part 6

☐ Piston #1 ☐ Piston #2

☐ Door #1 ☐ Door #2

☐ A START block and a GOAL block

3 · AND means BOTH · OR means EITHER 🔍

AND is true only when **both** things are true. **OR** is true when **at least one** is true.

```
if block on left AND redstone ahead:  
    turn left
```

There is a block on the left, but NOT redstone ahead. Does it turn? Circle one: yes · no


```
if block ahead OR redstone on right:  
    open it
```

There is a block ahead, but NO redstone on the right. Does it open it? Circle one: yes · no

4 · Write the Solution

Fill in the code to get the agent through your maze. Use a loop so it checks every step. You need **4 AND** checks and **1 OR** check in total.

```
while True:
    agent.move(FORWARD, 1)

    # turn only when BOTH are true (AND)
    if agent.detect(BLOCK, LEFT) and agent.detect(REDSTONE, AHEAD):
        agent.turn(LEFT_TURN)

    # a door OR a piston gate (OR)
    elif agent.detect(BLOCK, AHEAD) or agent.detect(REDSTONE, RIGHT):
        agent.interact(AHEAD)
        agent.move(FORWARD, 1)
```

Write your own checks for your maze here. Put **# AND** or **# OR** next to each one.

How many AND checks did you write? How many OR checks?

5 · Test It and Finish 📷

Type `run` and watch the agent. If it gets stuck, fix it and run again. Use `rl` to send it back to start. Keep going until the agent reaches the **GOAL** on its own.

When it finishes, send a photo or video of the agent at the goal and explain. Use these sentence starters — write 2 or 3 sentences.

My maze has 6 parts. The hardest part to build was ...

I put my pistons in ... and my doors in ...

I used AND to make the agent ... and OR to ...

6 · Present Your Maze 📹

Take a video on your phone (or a parent's phone) and show off **your** maze like a game designer. Try to use these words: **part, piston, door, AND, OR.**

1. Show the whole maze and point out the 6 parts.
2. Show each piston and each door.
3. Type **run** and let the agent reach the goal.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.